

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Powers, Roots & Standard Form

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

POWERS, ROOTS & STANDARD FORM · CH1 · L4



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The Map

This lesson collapses Powers, Roots & Standard Form into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Handle negative and fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Calculate then return to standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Difference

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Write this in standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 30 website questions, 30 checked solutions, 3 local PDFs read.

01 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out 5610710281043
 $3 \dots$ times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

02 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM2H Q13

Use index laws

This is a reasoning step: write one clear sentence, then continue the calculation.

Solve inequality

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 7.29×10^{242}

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Handle negative and fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out 5610710281043
 $3 \dots$ times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

04 Calculate then return to standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.357 \times 10^5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q8

The greatest coastline is the

$$1.357 \times 10^5$$

Convert standard form

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) Pacific Ocean, (b) 9.393×10^4 km

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

8 The table gives the length of the coastline, in kilometres, of each of five oceans.

Ocean	Length of coastline (km)
Arctic	4.539×10^4
Atlantic	1.119×10^5
Pacific	1.357×10^5
Indian	6.653×10^4
Southern	1.797×10^4

(a) Which ocean has the greatest length of coastline? (1) (b) Calculate the difference between the length of the Atlantic Ocean's coastline and the length of the Southern Ocean's coastline. Give your answer in standard form. km (2)

YOUR SOLUTION

05 Difference

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.06 \times 10^8 = 106000000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q10

Atlantic surface area

$$1.06 \times 10^8 = 106000000$$

Indian surface area

$$6.86 \times 10^7 = 68600000$$

Finish: (a) 3.74×10^7 , (b) 11

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	156 times 10 ⁸
Indian	6.86 times 10 ⁷
Southern	2.03 times 10 ⁷
Arctic	1.41 times 10 ⁷
Atlantic	1.06 times 10 ⁸

(a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean. Give your answer in standard form. square kilometres

(2) The surface area of the Pacific Ocean is k times the surface area of the ...

YOUR SOLUTION

06 Write this in standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 4.2 \times 10^{-24}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q8

Convert standard form

$$a = 4.2 \times 10^{-24}$$

Convert standard form

$$ab = (4.2 \times 3) \times 10^{-24+145}$$

Finish: 1.26×10^{122}

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

8 a = 4.2 times 10^{-24} b = 3 times 10145 Work out the value of a times b Give your answer in standard form.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Handle negative and fractional powers
"work out", "show", "hence", a familiar exam phrase	Calculate then return to standard form
"work out", "show", "hence", a familiar exam phrase	Difference
"work out", "show", "hence", a familiar exam phrase	Write this in standard form

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.